

REPUBLIQUE DU CAMEROUN
Paix-Travail-Patrie

MINISTÈRE DE L'ENSEIGNEMENT SUPÉRIEUR

COMMISSION NATIONALE D'ORGANISATION DE L'EXAMEN
HIGHER NATIONAL DIPLOMA EXAM (HND)

REPUBLIC OF CAMEROON
Peace-Work-Fatherland

MINISTRY OF HIGHER EDUCATION

NATIONAL COMMISSION FOR THE ORGANISATION OF
HIGHER NATIONAL DIPLOMA EXAM (HND) EXAM

National Exam of Higher National Diploma 2023 Session

Speciality/Option: SOFTWARE ENGINEERING (SWE)

Paper : CASE STUDY

Duration : 6 hours

Credit : 14

ANSWER ALL QUESTIONS

SECTION A: ALGORITHM AND PROGRAMMING (50 marks)

Question 1 Algorithms (10 marks)

Consider the following loop:

Declare Number As Integer

Set Number = 1

Repeat

Write 2 * Number

Set Number = Number + 1

Until Number > 10

Work required:

- Is this loop a pre-test loop or a post-test loop? Why?
- List the statements in the body of the loop.
- What is the test condition for this loop?
- What is the output of this loop?
- Draw the flowchart of this loop

2 marks

2 marks

2 marks

2 marks

2 marks

Question 2 Pseudocode (25 marks)

Consider the following pseudocode

Declare Q[10][10] As Integer

Declare R, C, As Integer

For (R = 1; R <= 3; R++)

For (C = 1; C <= 3; C++)

If R == C Then

Set Q[R][C] = 1

Else

Set Q[R][C] = 0


```

    End If
  End For(C)
End For(R)
For (R = 1; R <= 3; R++)
  For (C = 1; C <= 3; C++)
    Write Q[R][C]
  End For(C)
End For(R)

```

Work required:

- What is the output of code corresponding to the following pseudocode? **10 marks**
- Implement the previous pseudocode in the language you master most. **15 marks**

Question 3 Object oriented programming (15 marks)

- What are the two major components of an object? **(2 marks)**
- What is the relationship between a class and objects in that class? **(2 marks)**
- What is the difference between public and private class members? **(2 marks)**
- What is data hiding? Give an example of why it is important in an OOP program? **(2 marks)**
This is securing data in the class
- Define a class called HND, including a constructor, that has one attribute named Value (of type Float) and two methods as follows:
 - SetValue() is a subprogram that imports a value of the attribute Value from the main program. **(2 marks)**
 - GetValue() is a function that returns the value of the attribute Value to the main program. **(2 marks)**
- What is a constructor? Why are constructors used? **(3 marks)**

SECTION B: DATABASE DEVELOPMENT AND ADMINISTRATION (20 marks)

Question 4 (7 marks)

- What is meant by normalization in database? **(1mark)**
 - State and explain the different norm form. **(3marks)**
 - For the figure below state whether it is normalized or not. If not, state the norm form and normalize the table. **(3marks)**

DEPARTMENT			
DNAME	<u>DNUMBER</u>	DMGRSSN	DLOCATIONS
Research	5	333445555	{Bellaire, Sugarland, Houston}
Administration	4	987654321	{Stafford}
Headquarters	1	888665555	{Houston}

Question 5 (13 marks)

- What is functional dependency (1mark)
- Give examples of DDL, DCL and DML as used in SQL (2marks)
- The questions below concerns the table below
 - What is the cardinality and degree of the relation (2marks)
 - List the attributes and choose a suitable primary key attribute for the relation. (2marks)
 - Write an SQL command to create the table below. The table name should be *student_info* (1mark)
 - Write an SQL command to select all students with gpa greater than 2.7 (1.5marks)
 - Write an SQL command to select names of all the students (1mark)
 - Write an SQL command to add a fifth record to the table with *ID ST002*, *Name John*, *gpa 3.0* (1.5marks)
 - Write an SQL command to change the gpa of Suzy from 3.2 to 3.4 (1mark)

SECTION C: WEB DESIGN (15 marks)

Question 6

Write a HTML code to generate following output: (15 marks)

Country	Population (In Crores)	
	1998	1999
INDIA	85	90
	1999	100
	2000	
USA	30	35
	1999	40
	2000	
UK	25	30
	1999	35
	2000	

SECTION D: NETWORKING (15 marks)

Question 7

You have been provided with 2 computers and one computer that will act as the central computer to network and ensure that they should communication and share resources from a central computer.

work required

- What is a network? (1 mark)
- What is a computer network? (2 marks)
- What is the most appropriate OSs you think can be installed in both computers and the central computer so as to ensure that proper file sharing and resource access should be ensured? (2 + 1 = 3 marks)
- List all the various networking tools you will need in course of this activity to ensure that these two computers communicate with the central machine. (3 marks)
- Which other name can be given to the central machine? (2 marks)
- Which other name can be given to the two computers connected to the central machine? (1 mark)
- Which stand will you use to prepare your cables and why? (1 mark)
- LIST all the necessary configurations you think should be put in place and the various ping commands to ensure that the computers are on a network and can communicate and share resources (2 marks)